

Spectrum of radiation from two-level atoms under intense bichromatic excitation

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We develop a theoretical framework for calculating the spectrum of resonance fluorescence emitted by two-level atoms that are excited by a strong bichromatic field. The dressed states for the two-level system that is interacting with a fully modulated field are identified, and the spectral features are interpreted in terms of transitions among these dressed states.

INTRODUCTION

Determining the characteristics of the radiation emitted by an atom interacting with an intense field is one of the key problems in quantum optics. In an early study Mollow¹ pointed out that the spectrum of the fluorescence is considerably different from the spectrum that is expected for weak monochromatic excitation.^{2,3} It was later shown that the spectrum for a strong monochromatic field could easily be interpreted in terms of the dressed states⁴ of the system.

More recently Zhu *et al.*⁵ experimentally demonstrated that the fluorescence spectrum for the case of bichromatic excitation⁶⁻¹² differs qualitatively from the characteristic triplet spectrum that is observed for the case of strong monochromatic excitation. In this paper we show that the bichromatic-field-induced fluorescence spectrum can be calculated by solving the coupled equations for the dipole-dipole correlation functions with continued-fraction methods. We find that the spectrum consists of multiple peaks whose spacing is independent of the Rabi frequency and that for on-resonance excitation the spacing is equal to half of the frequency difference of the two fields. These results are interpreted in terms of the dressed states of the system and the possible transitions among them. We also point out that a significant fraction of the total scattered power is contained in the elastic part of the spectrum even for large values of the Rabi frequency.

CALCULATION OF THE EMISSION SPECTRUM BY CONTINUED-FRACTION METHODS

Consider a two-level atom with transition frequency ω_0 and upper and lower states denoted by $|1\rangle$ and $|0\rangle$, respectively, that is driven by a bichromatic field with components ω_1 and ω_2 . Let S^+ and S^- be the spin-1/2 angular-momentum operators for the dipole moment and for the inversion,¹³ respectively. In a frame that is rotating at frequency ω_2 the optical Bloch equations, generalized for the case of bichromatic-field excitation, can be

written as

$$\frac{\partial \phi}{\partial t} = \mathbf{M}\phi + [\mathbf{M}_+ \exp(-i\Omega t) + \mathbf{M}_- \exp(i\Omega t)]\phi + \mathbf{I}, \quad (1)$$

where $\Omega = \omega_1 - \omega_2$ and where the column matrix ϕ has components $\phi_1 = \langle S^+(t) \rangle$, $\phi_2 = \langle S^-(t) \rangle$, and $\phi_3 = \langle S^z(t) \rangle$. The nonzero elements of the other matrices are

$$M_{11} = M_{22}^* = (i\Delta_2 - 1/T_2), \quad M_{33} = -1/T_1,$$

$$M_{13} = M_{23}^* = -M_{31}^*/2 = -M_{32}/2 = ig_2^*/2,$$

$$(M_+)_{23} = -(M_+)_{31}/2 = -ig_1/2,$$

$$(M_-)_{13} = -(M_-)_{32}/2 = ig_1^*/2, \quad I_3 = \eta/T_1,$$

where $\Delta_2 = \omega_0 - \omega_2$. The longitudinal and the transverse relaxation times are denoted by T_1 and T_2 , respectively, and η is the population inversion in the absence of an applied field. The Rabi frequencies for the two components of the field are denoted by g_1 and g_2 .

The resonance fluorescence spectrum is given in terms of the atomic dipole-dipole correlation function. Hence we introduce the correlation matrix $\Phi(t + \tau, t)$ with components

$$\Phi_1(t + \tau, t) = \langle S^+(t + \tau)S^-(t) \rangle - \langle S^+(t + \tau) \rangle \langle S^-(t) \rangle,$$

$$\Phi_2(t + \tau, t) = \langle S^-(t + \tau)S^-(t) \rangle - \langle S^-(t + \tau) \rangle \langle S^-(t) \rangle,$$

$$\Phi_3(t + \tau, t) = \langle S^z(t + \tau)S^-(t) \rangle - \langle S^z(t + \tau) \rangle \langle S^-(t) \rangle. \quad (2)$$

The inelastic spectrum is related to the correlation function Φ_1 . It can be shown by means of the quantum regression theorem that the correlation matrix Φ satisfies Eq. (1) with $\mathbf{I} = 0$ and the substitutions $\partial/\partial t \rightarrow \partial/\partial \tau$ and $\exp(\pm i\Omega t) \rightarrow \exp[\pm i\Omega(t + \tau)]$. This new equation of motion can be simplified by decomposition of the correlation matrix into slowly varying amplitudes that oscillate at the modulation frequency and at its harmonics. This decomposition is given by the relation

$$\Phi = \sum_{l=-\infty}^{+\infty} \Phi^{(l)} \exp[-il\Omega(t + \tau)]. \quad (3)$$

The evolution of these amplitudes is governed by

$$\frac{\partial \Phi^{(l)}}{\partial \tau} = (\mathbf{M} + i\Omega)\Phi^{(l)} + \mathbf{M}_+\Phi^{(l-1)} + \mathbf{M}_-\Phi^{(l+1)}. \quad (4)$$

We solve Eq. (4) by using Laplace transform techniques and denote the transformed correlation as $\hat{\Phi}^{(l)}(z)$, where $z = i(\omega - \omega_2)$. One can show that $\hat{\Phi}_1$ can be obtained from

$$\hat{\Phi}_1^{(l)}(z) = [\Phi_1^{(l)}(\tau = 0) + \frac{1}{2}ig_2^*\hat{\Phi}_3^{(l)} + \frac{1}{2}ig_1^*\hat{\Phi}_3^{(l+1)}]P_l. \quad (5)$$

The components $\hat{\Phi}_3^{(l)}$ and $\hat{\Phi}_3^{(l+1)}$ are determined from an inhomogeneous, three-term recursion relation given by

$$A_l\hat{\Phi}_3^{(l)} + B_l\hat{\Phi}_3^{(l-1)} + C_l\hat{\Phi}_3^{(l+1)} = D_l. \quad (6)$$

The coefficients that appear in the recursion relation are given by

$$\begin{aligned} A_l &= z + (1/T_1) - i\Omega + \frac{1}{2}|g_2|^2(P_l + Q_l) \\ &\quad + \frac{1}{2}|g_1|^2(P_{l-1} + Q_{l+1}), \\ B_l &= \frac{1}{2}g_1g_2^*(Q_l + P_{l-1}), \quad C_l = \frac{1}{2}g_1^*g_2(P_l + Q_{l+1}), \\ D_l &= ig_2P_l\Phi_1^{(l)}(\tau = 0) - ig_2^*Q_l\Phi_2^{(l)}(\tau = 0) \\ &\quad + ig_1P_{l-1}\Phi_1^{(l-1)}(\tau = 0) - ig_1^*Q_{l+1}\Phi_2^{(l+1)}(\tau = 0) \\ &\quad + \Phi_3^{(l)}(\tau = 0), \end{aligned}$$

where

$$\begin{aligned} P_l &= [z + (1/T_2) - i\Omega - i\Delta_2]^{-1}, \\ Q_l &= [z + (1/T_2) - i\Omega + i\Delta_2]^{-1}. \end{aligned}$$

The physical quantity of interest is the temporally stationary component of Φ_1 , which is denoted by $\Phi_1^{(0)}$ and can be obtained from Eq. (5) after substitution of Eq. (6). The solution of the recursion relation of Eq. (6) for the correlation amplitudes $\Phi_3^{(l)}$ is complex because of the inhomogeneities and is obtained with Green's-function techniques for recursion relations.¹⁴

The recursion relation of Eq. (6) requires the knowledge of the initial values of the correlations, which are given in terms of the single-time expectation values by

$$\begin{aligned} \Phi_1^{(l)}(\tau = 0) &= \exp(i\Omega\tau) (\frac{1}{2} + \frac{1}{2}\phi_3 - \phi_1\phi_2), \\ \Phi_2^{(l)}(\tau = 0) &= -\exp(i\Omega\tau)\phi_2\phi_3, \\ \Phi_3^{(l)}(\tau = 0) &= -\exp(i\Omega\tau)\phi_2(1 + \phi_3). \end{aligned} \quad (7)$$

We decompose the single-time expectation values into slowly varying amplitudes given by

$$\phi = \sum_{r=-\infty}^{+\infty} \phi^{(r)} \exp(-ir\Omega\tau). \quad (8)$$

The slowly varying amplitude for the population inversion $\phi_3^{(r)}$ can be obtained from the recursion relation, Eq. (6), with $l = r, z = 0$, and with D_l replaced by $\eta\delta_{r,0}/T_1$, which can be solved with standard continued-fraction methods. The other slowly varying amplitudes are then obtained through the relations

$$\begin{aligned} \phi_1^{(r)} &= \frac{1}{2}[ig_2^*\phi_3^{(r)} + ig_1^*\phi_3^{(r+1)}]p_r, \\ \phi_2^{(r)} &= -\frac{1}{2}[ig_2\phi_3^{(r)} + ig_1\phi_3^{(r-1)}]q_r, \end{aligned} \quad (9)$$

where $p_r = P_r(z = 0)$ and $q_r = Q_r(z = 0)$.

The inelastic spectrum $S_{in}(\omega)$ is obtained from $\hat{\Phi}_1$ according to

$$S_{in}(\omega) = \hat{\Phi}_1^{(0)}[i(\omega - \omega_2)] + \text{c.c.} \quad (10)$$

The elastic part of the spectrum is obtained from $\langle S^+(t + \tau) \rangle \langle S^-(t) \rangle$, which is given by

$$S_{el}(\omega) = \sum_{j=-\infty}^{+\infty} \delta(\omega - \omega_2 + j\Omega) |\phi_1^{(j)}|^2. \quad (11)$$

With the above formulation the elastic and inelastic contributions to the fluorescence spectrum can be calculated under a wide range of parameters of the bichromatic field.

We calculate fluorescence spectra by using values of the parameters that correspond to the situation studied experimentally by Zhu *et al.*,⁵ i.e., $(\omega_1 - \omega_0)T_2 = -(\omega_2 - \omega_0)T_2 = 10$, $\Omega T_2 = 20$, $(\omega_1 + \omega_2)/2 = \omega_0$, $|g_1| = |g_2| \equiv |g|$, $n = -1$, and $T_2 = 2T_1$. In the experiment the spectrum was measured for various values of the Rabi frequency $|g|$. The corresponding fluorescence spectra that are predicted by Eqs. (10) and (11) are shown in Fig. 1. We note that the spectra are symmetric. It is seen that the elastic peaks occur at frequencies $\omega_0 \pm (k + 1/2)\Omega$, ($k = 0, 1, 2, \dots$), as predicted by Eq. (11). The intensity of a given elastic peak depends on $|\phi_1^{(r)}|^2$, and it is seen

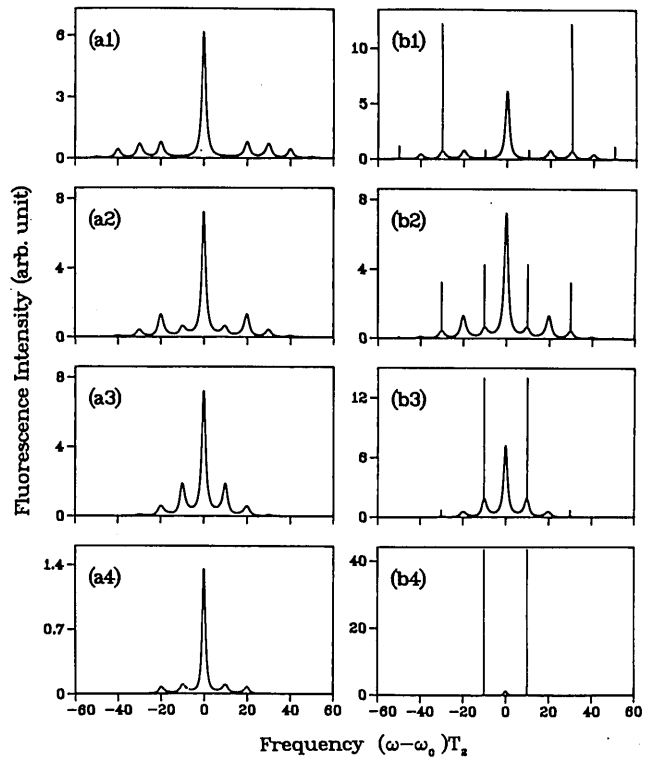


Fig. 1. Theoretical fluorescence spectra produced by bichromatic-field excitation under the conditions $\Omega T_2 = 20$, $\Delta_2 T_2 = 10$, and $T_2 = 2T_1$. The spectra, from bottom to top, are for increasing values of the resonant Rabi frequency $g = |g_1| = |g_2|$ of the field components. gT_2 is (a1) 20, (a2) 15, (a3) 10, and (a4) 5. The curves in the left-hand column give the inelastic contributions to the fluorescence spectra, whereas the curves in the right-hand column show both elastic and inelastic contributions. The spectra are convolved with a Lorentzian transmission function with a linewidth $0.01/T_2$. The vertical axes of the figures are labeled with consistent units so that the fluorescence intensities can be compared. The average frequency $(\omega_1 + \omega_2)/2$ of the bichromatic field is equal to the atomic frequency ω_0 .

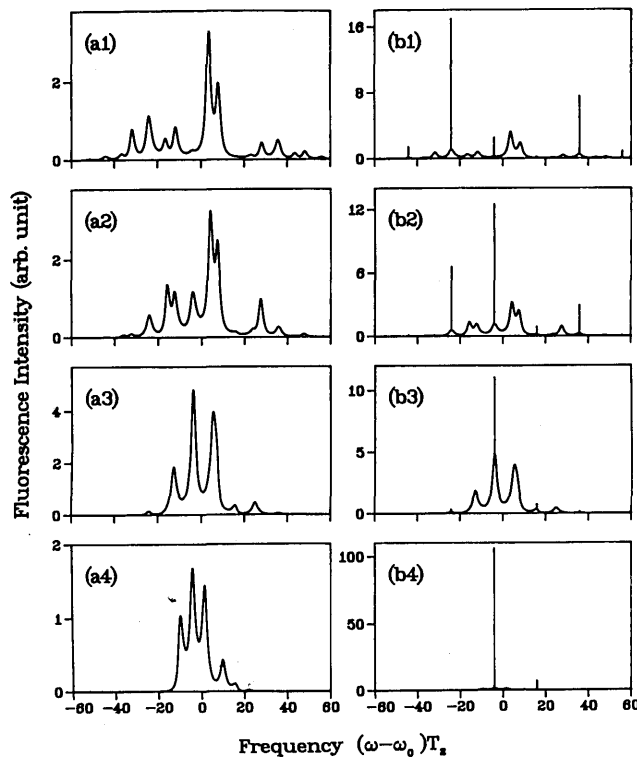


Fig. 2. Calculated fluorescence spectra for bichromatic excitation. The parameters are the same as those in Fig. 1 except $(\omega_1 + \omega_2)/2 - \omega_0 = 6/T_2$, corresponding to off-resonance excitation. The relative intensity units are the same as those of Fig. 1.

that some elastic peaks are greatly enhanced for particular values of the Rabi frequency. The inelastic peaks occur at the g -independent frequencies $\omega_0 \pm k\Omega/2$ ($k = 0, 1, 2, \dots$). Although the emission frequencies are g independent, the number of peaks in the spectrum increases as the Rabi frequency increases. We determine the width of the peaks directly from the plots and find that the linewidth (FWHM) of the central inelastic peak at ω_0 is approximately equal to $1/T_1$ and that the linewidth for the other inelastic peaks is approximately equal to $3/(2T_1)$. Our results are consistent with the numerical calculations of Newbold and Salamo,¹⁵ who calculated the spectrum by direct integration of the equations of motion for the correlation functions.

An interesting feature in the experimentally measured spectra⁵ is that the spectral peaks at frequencies $\omega_0 \pm (k + 1/2)\Omega$ ($k = 0, 1, 2, \dots$) were narrower than the peaks at frequencies $\omega_0 \pm k\Omega$. We believe that the apparent narrowness of the peaks at frequencies $\omega_0 \pm (k + 1/2)\Omega$ results from the presence of both inelastic and narrow-linewidth elastic contributions to the fluorescence. Quantitative comparison of our theory and the experiment of Zhu et al.⁵ requires detailed knowledge of experimental parameters that are not directly considered in our theory, such as the homogeneity of the driving field, the residual Doppler width, and the instrumental linewidth.

In Fig. 2 we plot the calculated spectra for the bichromatic excitation with the same parameters as those in Fig. 1 except $(\omega_1 + \omega_2)/2 - \omega_0 = 6/T_2$, corresponding to off-resonance excitation. The spectra are markedly asymmetric and apparently contain more peaks than in Fig. 1.

The calculated fluorescence spectra of Fig. 1 qualitatively demonstrate the importance of elastic scattering.

We quantify these results by calculating the total fluorescence intensity I_{tot} , the total elastic intensity $I_{\text{tot}}^{\text{el}}$, and the fluorescence intensity of some individual elastic peaks I_s^{el} as functions of the Rabi frequency. Results of these calculations are shown in Fig. 3, where for the purpose of comparison we plot the analogous quantities that are characteristic of resonant monochromatic excitation. In Fig. 3(a) we plot the total fluorescence intensity $I_{\text{tot}} \propto (1 + \langle S^2 \rangle)/2$ and find, in agreement with previous studies,⁷⁻¹¹ that it oscillates as a function of the Rabi frequency g . The peaks of I_{tot} have been referred to as Rabi subharmonic resonances.

Also plotted in Fig. 3(a) is the total elastic intensity, which is given by

$$I_{\text{tot}}^{\text{el}} \propto \sum_{m=-\infty}^{+\infty} |\phi_1^{(m)}|^2. \quad (12)$$

One sees that in the bichromatic case elastic scattering remains significant for large values of the Rabi frequency and that the total elastic intensity also displays resonances. In Fig. 3(b) we plot the intensities of the first three elastic peaks, which occur at frequencies $\omega_0 \pm s\Omega/2$ ($s = 1, 3, 5$). Comparing Figs. 3(a) and 3(b), one sees that clearly the k th minimum in I_{tot} occurs at the same Rabi frequency as the maximum in the intensity of the $(2k + 1)$ th elastic peak for $k \geq 1$. The oscillations that are observed in the bichromatic case are completely absent in the case of resonant monochromatic excitation.

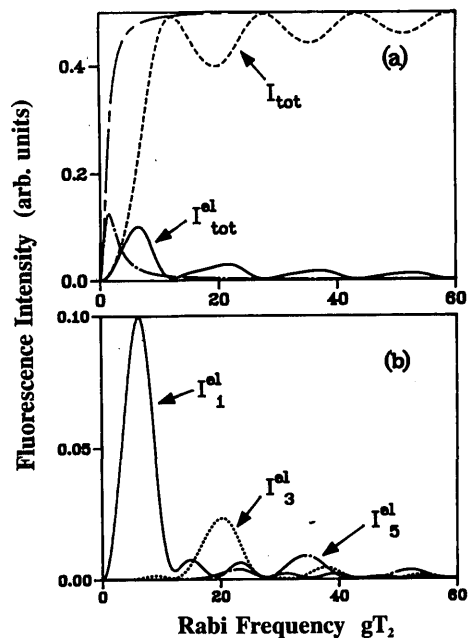


Fig. 3. (a) Contributions to the fluorescence intensity versus the Rabi frequency g . The dashed curve represents the total scattering intensity I_{tot} , and the solid curve represents the total elastic scattering intensity $I_{\text{tot}}^{\text{el}}$ for bichromatic excitation, as described in Fig. 1. The long-and-short-dashed curve and the dot-dashed curve represent the corresponding quantities for monochromatic excitation. (b) Elastic scattering intensity I_s^{el} for peaks at frequencies $\omega_0 \pm s\Omega/2$ (summed over + and -) as a function of the Rabi frequency g for the case of bichromatic excitation. Solid curve, elastic scattering intensity I_1^{el} ; dotted curve, elastic scattering intensity I_3^{el} ; dot-dashed curve, elastic scattering intensity I_5^{el} .

INTERPRETATION OF THE SPECTRUM IN TERMS OF THE DRESSED STATES

We now develop a quantum-theoretical description of the interaction of the atom with a fully amplitude-modulated, two-frequency-component field. For $(\omega_1 + \omega_2)/2 = \omega_0$, the semiclassical Hamiltonian in the interaction picture for the atom and the fully modulated field is given by

$$H_1 = -g/2[\exp(-i\Omega t/2) + \exp(i\Omega t/2)](S^+ + S^-), \quad (13)$$

and the fully quantized version of this Hamiltonian is given by

$$H = \frac{1}{2}\Omega b^\dagger b - g_0(b + b^\dagger)S^x, \quad (14)$$

where $S^x = (S^+ + S^-)/2$. The quantization is carried out for the field with modulation frequency $\Omega/2$. If the mode b is in a coherent state $|\beta\rangle$, the coupling constant g_0 is related to the Rabi frequency g of the optical field by $g_0 = g/\beta = g/N^{1/2}$, where N is the expected number of modulated photons.¹⁶ The photon content of the modulated field can be obtained easily by noting that the Hamiltonian given in Eq. (13) is essentially that for a displaced harmonic oscillator whose eigenstates are well known.¹⁷

Using the displacement operator $D(\alpha) = \exp(\alpha b^\dagger - \alpha^* b)$,¹⁸ the eigenstates of the Hamiltonian can be shown to be¹⁹

$$|\Psi_\pm^{(n)}\rangle = D(\pm g_0/\Omega)|n\rangle|\psi_\pm\rangle, \quad (15)$$

where $|\psi_\pm\rangle = 1/\sqrt{2}(|1\rangle \pm |0\rangle)$ and $|n\rangle$ is the number state for the modulated field. Written in terms of the eigenstates $|\Psi_\pm^{(n)}\rangle$, the Hamiltonian with manifest eigenvalues becomes

$$H|\Psi_\pm^{(n)}\rangle = \left(\frac{n\Omega}{2} - \frac{g_0^2}{2\Omega}\right)|\Psi_\pm^{(n)}\rangle. \quad (16)$$

The matrix elements of D are given by

$$\langle m|D(\alpha)|n\rangle = \exp\left(-\frac{\alpha^2}{2}\right) \sum_q \frac{(-1)^q (\alpha)^{2q+m-n} (n! m!)^{1/2}}{q!(q+m-n)!(n-q)!}, \quad (17)$$

where we take α to be real. The matrix elements, Eq. (17), can be simplified in the semiclassical limit $\alpha \rightarrow 0$, $n \rightarrow \infty$, $\alpha\sqrt{n} \rightarrow \text{constant}$ and are approximately given by

$$\langle m|D(\alpha)|n\rangle = J_{m-n}(2\alpha\sqrt{n}), \quad (18)$$

where J_n is the Bessel function of order n .

The spectral features in the resonance fluorescence spectra can be interpreted as arising from transitions among the dressed states²⁰ given in Eq. (15). The transition matrix elements are

$$\langle \Psi_\pm^{(m)}|S^+|\Psi_\pm^{(n)}\rangle = \pm \frac{1}{2}\delta_{mn}, \quad (19)$$

$$\langle \Psi_\pm^{(m)}|S^+|\Psi_\mp^{(n)}\rangle = \pm \frac{1}{2}\langle m|D(\pm 2g_0/\Omega)|n\rangle. \quad (20)$$

The matrix elements of Eq. (19) contribute to the central peak at $\omega = \omega_0$, whereas the matrix elements of Eq. (20) lead to peaks at $\omega = \omega_0 + (m - n)\Omega/2$. In the semiclassical limit the matrix elements of Eq. (20) are given approximately by

$$\langle \Psi_\pm^{(m)}|S^+|\Psi_\mp^{(n)}\rangle = \pm \frac{1}{2}J_{m-n}(\pm 4g/\Omega). \quad (21)$$

The width of the transitions can be calculated from the decay of the density matrix elements in the basis, Eq. (15). For coherent transitions (elastic scattering), interference between transition amplitudes leads to the suppression of some spectral features. For example, the elastic scattering at $\omega_0 + (m - n)\Omega/2$ will be determined by

$$\begin{aligned} &\langle \Psi_-^{(m)}|S^+|\Psi_+^{(n)}\rangle + \langle \Psi_+^{(m)}|S^+|\Psi_-^{(n)}\rangle \\ &= \frac{1}{2} \left\langle m \left| \left[D\left(\frac{2g_0}{\Omega}\right) - D\left(-\frac{2g_0}{\Omega}\right) \right] \right| n \right\rangle, \end{aligned} \quad (22)$$

which is zero when $m - n$ is even. We have verified that the matrix elements of Eqs. (19) and (20) correctly predict the intensity of the spectral features shown in Fig. 1.

It should be noted that for a fully amplitude-modulated field the field amplitude can vary as $\cos(\Omega t/2)$, a bichromatic field, or as $[1 + \cos(\Omega t/2)]$, a three-frequency field. The later case was analyzed by Blind *et al.*,²⁰ who showed that the atom-field eigenstates are the same as those shown in Eq. (15). In contrast, the eigenvalues for three-frequency excitation are equal to $\pm \frac{1}{2}g \pm \frac{1}{2}n\Omega - g_0^2/2\Omega$, which depend on the Rabi frequency and hence are dramatically different from those for bichromatic excitation [see Eq. (16)]. Consequently the elastic scattering for three-frequency excitation occurs at frequencies $\omega_0 \pm n\Omega/2$, whereas the inelastic scattering occurs at $\omega_0 \pm g \pm n\Omega/2$. This result shows that the fluorescence spectrum of two-level atoms that are excited by a fully amplitude-modulated field can have quite different characteristics, which are dependent on the specific form of the field. However, the variation of the total scattered intensity and the total elastic intensity as functions of the Rabi frequency are quite similar to the results for a three-frequency field.

CONCLUSIONS

In conclusion, we have developed a theoretical framework with a continued-fraction method, which is used to calculate the fluorescence spectrum of two-level atoms under strong bichromatic excitation. We have derived the analytical expression for the dressed states of the system that consists of two-level atoms interacting with a fully amplitude-modulated, bichromatic field. The spectra can be interpreted in terms of the transitions among these dressed atom-field states.

Note added in proof: It has come to our attention that H. Freedhoff and Z. Chen,²¹ Kumari and Tewari,²² and Kumari²³ recently reported methods for calculation of the resonance fluorescence spectrum of a two-level atom in a bichromatic field. Some of the results presented in this paper appear to differ from those reported by Freedhoff and Chen.

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REFERENCES AND NOTES

1. B. R. Mollow, Phys. Rev. **188**, 1969 (1969); see also A. I. Burshtein, Sov. Phys. JEPT **21**, 567 (1965); **22**, 939 (1966) [J. Exp. Theor. Phys. (U.S.S.R.) **48**, 850 (1965); **49**, 1362 (1965)].
2. W. Heitler, *The Quantum Theory of Radiation*, 3rd ed. (Oxford U. Press, London, 1960), pp. 196–204.
3. The effect of nonmonochromaticity on the Mollow spectra has been investigated by G. S. Agarwal, Phys. Rev. Lett. **37**, 1383 (1976); J. H. Eberly, Phys. Rev. Lett. **37**, 1387 (1976).
4. C. Cohen-Tannoudji and S. Reynaud, J. Phys. B **10**, 345 (1977).
5. Y. Zhu, Q. Wu, A. Lezama, D. J. Gauthier, and T. W. Mossberg, Phys. Rev. A **41**, 6547 (1990).
6. Extensive research on emission and absorption in bichromatic fields has been done. For example, Refs. 7 and 8 (and references therein) examine in detail the total fluorescence and absorption as a function of one of the frequencies of the bichromatic field.
7. G. S. Agarwal and N. Nayak, J. Opt. Soc. Am. B **1**, 164 (1984); Phys. Rev. A **33**, 391 (1986); J. Phys. B **19**, 3385 (1986).
8. P. Thomann, J. Phys. B **13**, 1111 (1980).
9. S. Chakmakjian, K. Koch, and C. R. Stroud, Jr., J. Opt. Soc. Am. B **5**, 2015 (1988).
10. S. P. Goreslavsky, N. B. Delone, and V. P. Krainov, J. Phys. B **13**, 2659 (1980).
11. W. M. Ruyten, Opt. Lett. **14**, 506 (1989); Phys. Rev. A **40**, 1447 (1989); J. Opt. Soc. Am. B **6**, 1796 (1989).
12. Rich spectra associated with doubly amplitude-modulated excitation fields have been predicted by M. Wilkens and K. Rzazewski, Phys. Rev. A **40**, 3164 (1989).
13. The expectation values of the spin operators are given in terms of the expectation values of the density operator through the relations $\langle S^-(t) \rangle = \langle 1|\rho|0 \rangle \exp(i\omega_2 t)$, $\langle S^+(t) \rangle = \langle 0|\rho|1 \rangle \exp(-i\omega_2 t)$, and $\langle S^z(t) \rangle = \langle 1|\rho|1 \rangle - \langle 0|\rho|0 \rangle$.
14. See H. Risken, *The Fokker-Planck Equation* (Springer-Verlag, Berlin, 1984), Chap. 9, for an example of how Green's-function techniques can be applied to solve inhomogeneous three-term recursion relations.
15. M. A. Newbold and G. J. Salamo, Phys. Rev. A **22**, 2098 (1980).
16. Note that the number of modulated photons is equal to the number of optical photons, since the quantization is performed in the interaction picture. See, for example, B. Blind, P. R. Fontana, and P. Thomann, J. Phys. B **13**, 2717 (1980).
17. P. Carruthers and M. M. Nieto, Am. J. Phys. **33**, 537 (1965).
18. R. J. Glauber, Phys. Rev. **131**, 2766 (1963).
19. N. Polansky and C. Cohen-Tannoudji, J. Phys. (Paris) **26**, 409 (1965).
20. B. Blind, P. R. Fontana, and P. Thomann, J. Phys. B **13**, 2728 (1980), present approximate analytical results for the fluorescence spectra in an amplitude-modulated, three-frequency field under the assumption that the strength of the driving field is large enough to saturate the optical transition. They also assume that the carrier frequency of the driving field is resonant with the atomic frequency.
21. H. Freedhoff and Z. Chen, Phys. Rev. A **41**, 6013 (1990).
22. M. K. Kumari and S. P. Tewari, Phys. Rev. A **41**, 5273 (1990).
23. M. K. Kumari, Ph.D. dissertation (University of Hyderabad, Hyderabad, 1990).